APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

Correlation Regime Model

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Owner	Dr. Yuki Tanaka (Quant Researcher)
Validator	Dr. Samuel Achebe (Model Validation Officer)
Status	In Validation
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Regulatory Framework: Basel III Annex 4, SR 11-7

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Contents

1 Model Overview **3**
1.1 Purpose . . . 3
1.2 Business Use . . . 3

2 Theoretical Basis **3**
2.1 Regime Indicator . . . 3
2.2 Hysteresis State Machine . . . 3
2.3 Normal Regime Correlation Matrix (Σ_{NORMAL}) . . . 3
2.4 Stress Regime Correlation Matrix (Σ_{STRESS}) . . . 4

3 Mathematical Specification **4**

4 Implementation **4**

5 Validation **4**

6 Model Limitations **5**

7 Open Findings **5**

1 Model Overview

1.1 Purpose

The Correlation Regime Model (CRM) detects the prevailing cross-asset correlation regime (NORMAL vs STRESS) using a rolling equity-credit correlation indicator and applies the appropriate 6×6 correlation matrix to the VaR Monte Carlo Cholesky path generator (APEX-MDL-0001).

1.2 Business Use

- **VaR capital:** Regime-dependent correlation matrix is the primary input to Monte Carlo Cholesky factorisation in VaR (APEX-MDL-0001).
- **SVaR:** Stress regime matrix applied unconditionally for stressed capital (APEX-MDL-0002).
- **XVA:** Regime drives counterparty exposure profiles in PFE simulation.
- **Risk limit calibration:** Regime indicator triggers intra-day risk limit review when transitioning NORMAL $\rightarrow$ STRESS.

2 Theoretical Basis

2.1 Regime Indicator

The regime indicator is the 20-day rolling Pearson correlation between SPY daily log-returns (R_{eq}) and CDX IG 5Y daily log-returns (R_{cr}):

$$\rho_{\text{obs}}(t) = \text{Corr}\big(R_{\text{eq}}(t-19:t),\ R_{\text{cr}}(t-19:t)\big)$$

Under normal market conditions, equity returns and investment-grade credit spreads are negatively correlated ($\rho \approx -0.20$). Under stress, they co-move strongly in the negative direction ($\rho \approx -0.80$), reflecting the flight-to-quality and credit-equity dislocation phenomenon.

2.2 Hysteresis State Machine

To avoid regime flickering, a hysteresis band $[0.40, 0.65]$ is applied:

$$\text{Regime}(t) = \begin{cases} \text{STRESS} & \text{if } \rho_{\text{obs}}(t) > 0.65 \\ \text{NORMAL} & \text{if } \rho_{\text{obs}}(t) < 0.40 \\ \text{Regime}(t-1) & \text{otherwise (hysteresis band)} \end{cases}$$

Note: the indicator uses $|\rho_{\text{obs}}|$ for symmetric detection; the sign convention above corresponds to the absolute value of the equity-credit correlation.

2.3 Normal Regime Correlation Matrix (Σ_{NORMAL})

6×6 matrix across asset classes [EQ, FX, IR, CR, CMDTY, VOL]:

$$\Sigma_{\text{NORMAL}} = \begin{pmatrix} 1.00 & 0.30 & -0.20 & -0.20 & 0.20 & -0.60 \\ 0.30 & 1.00 & -0.10 & -0.10 & 0.15 & -0.30 \\ -0.20 & -0.10 & 1.00 & 0.50 & -0.10 & 0.10 \\ -0.20 & -0.10 & 0.50 & 1.00 & -0.15 & 0.15 \\ 0.20 & 0.15 & -0.10 & -0.15 & 1.00 & -0.20 \\ -0.60 & -0.30 & 0.10 & 0.15 & -0.20 & 1.00 \end{pmatrix}$$

2.4 Stress Regime Correlation Matrix (Σ_{STRESS})

Calibrated to 2008 GFC peak stress (Oct–Dec 2008):

$$\Sigma_{\text{STRESS}} = \begin{pmatrix} 1.00 & 0.60 & -0.40 & -0.80 & 0.50 & -0.90 \\ 0.60 & 1.00 & -0.20 & -0.50 & 0.35 & -0.60 \\ -0.40 & -0.20 & 1.00 & 0.70 & -0.30 & 0.30 \\ -0.80 & -0.50 & 0.70 & 1.00 & -0.40 & 0.60 \\ 0.50 & 0.35 & -0.30 & -0.40 & 1.00 & -0.50 \\ -0.90 & -0.60 & 0.30 & 0.60 & -0.50 & 1.00 \end{pmatrix}$$

Key stressed pairs: $\rho_{\text{EQ,CR}} = -0.80$, $\rho_{\text{EQ,VOL}} = -0.90$, $\rho_{\text{FX,CR}} = -0.50$.

3 Mathematical Specification

Parameter	Value	Notes
Rolling window	20 business days	$\approx$ 1 calendar month
STRESS trigger	$\lvert\rho_{\text{obs}}\rvert > 0.65$	Hysteresis upper bound
NORMAL trigger	$\lvert\rho_{\text{obs}}\rvert < 0.40$	Hysteresis lower bound
Hysteresis band	$[0.40, 0.65]$	Prevents flickering
Equity proxy	SPY daily log-returns	Observed from market feed
Credit proxy	CDX IG 5Y daily spread changes	Computed from counterparty registry

4 Implementation

Code location: `infrastructure/risk/correlation_regime.py`
Class: `CorrelationRegimeModel`

Key methods:

- `detect_regime(eq_returns, cr_returns)` — returns `RegimeType.NORMAL` or `RegimeType.STRESS`.
- `get_correlation_matrix(regime)` — returns 6×6 numpy array.
- `cholesky(regime)` — pre-computed Cholesky factor L such that $LL^\top = \Sigma_{\text{regime}}$.

Integration: `VaRCalculator.calculate()` calls `detect_regime()` daily. `VaRCalculator.calculate_stress` uses `STRESS` matrix unconditionally.

5 Validation

Correlation matrix validation: Both Σ_{NORMAL} and Σ_{STRESS} confirmed positive semi-definite by Cholesky decomposition (no numerical failures).

Backtesting: Regime detection applied retrospectively to 2008 GFC data; model enters STRESS regime within 3 days of Lehman Brothers default (2008-09-15). Exits STRESS regime by 2009-07 (aligns with S&P 500 recovery).

VaR uplift: STRESS regime VaR 2.1× higher than NORMAL regime for current portfolio; consistent with empirical GFC loss data.

6 Model Limitations

1. **Single regime boundary:** Binary NORMAL/STRESS does not capture partial-stress states (e.g., single-sector stress); a 3-state model (NORMAL/ELEVATED/STRESS) would improve granularity.
2. **20-day rolling lag:** Regime detection has ∼10-day lag relative to instantaneous correlation shift; rapid stress onset (e.g., flash crash) may not trigger in time.
3. **US-centric proxies:** SPY and CDX IG are US instruments; European stress (Gilt Crisis 2022) may not trigger the indicator until contagion reaches US markets.
4. **Proxy quality:** CDX IG 5Y spread change is computed from counterparty registry CDS spreads, not a traded index; basis risk between proxy and market CDX not quantified.

7 Open Findings

ID	Severity	Description	Status
CRM-F1	**Major**	Single regime boundary: binary NORMAL/STRESS misclassifies idiosyncratic sector stress; 3-state regime model recommended.	Open
CRM-F2	**Major**	20-day rolling window introduces ∼10-day lag; rapid stress onset may not trigger regime change within risk management response window.	Open
CRM-F3	**Minor**	US-centric proxies (SPY, CDX IG) may not detect European or EM-originated systemic stress events.	Open